

DSCI 351 Recommender Systems

Assignment 2: Model-based collaborative filtering

Dataset for assignment 2: movieLen_1M: ratings.dat, movies.dat

What to submit:

- A pdf file includes only the answers to questions in each step.
- A Python code file.

Please follow the steps, and answer questions in each step.

Step 1. Construct a rating matrix with the dataset provided, each row refers to user id, each column refers to movie id. Which movie has the highest number of known ratings? Provide movie id, movie title and known rating number.

Step 2. Split the rating matrix into X and Y. Where Y is the column of the movie with the highest number of known ratings in the previous step. X is the resting of the columns in rating matrix. After the split, for both X and Y, only keep the rows where have known rating in Y. What is the dimension of X now?

Step 3. Let's perform further cleaning with Y. The known ratings range from 1 to 5. For this model-based movie recommender, we would like to set a threshold, if the rating is higher than ($>$) the threshold, then we consider the movie can be recommended, otherwise ($\leq$), not recommended. Suppose the threshold is 3, change the value in Y based on such preference. What is the portion of movie to be recommended in Y? What is the sparsity of X? Then, impute 0 for unspecified ratings in X for modeling.

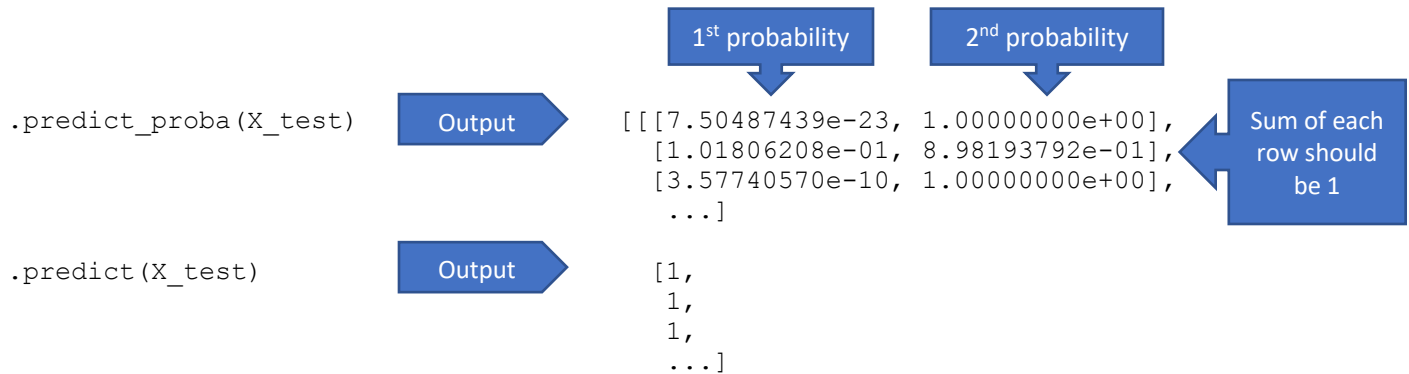
Step 4. Split the X and Y in train set (80%, `X_train`, `Y_train`) and test set (20%, `X_test`, `Y_test`), make sure your data split can be replicated. What is the dimension of `X_test` now?

Step 5. Now let us train a Naïve Bayes model. Look into the document from scikit-learn regarding [MultinomialNB](#) and [BernoulliNB](#), and observe your dataset, which Naïve Bayes method should be applied? Provide your explanation. With selected Naïve Bayes methods, with hyper-parameter setting as `alpha=1` and `fit_prior=True`, train with your train set (`X_train`, `Y_train`).

Step 6a. With trained Naïve Bayes model, make prediction with your test set (`.predict(X_test)`). What is the percentage of instances in `X_test` are correctly classified, when compare predictions to `Y_test`?

Step 6b. Now check the `.predict_proba(X_test)` output with trained Naïve Bayes model, each row has two probabilities (probability of the instance belong to class 0, and

probability of the instance belong to class 1), the sum of them should be equal to 1 in each row. Examine and compare the output to the predictions, what is threshold in 2nd probability result applied for classification prediction? If set the threshold as 0.6 to make the prediction, what is the percentage of instances in `X_test` are correctly classified now?



With 2nd probability higher than threshold, predict as class 1, otherwise, predict as class 0.

Step 7. Now use the splitted data from step 4, first apply

`sklearn.decomposition.TruncatedSVD(n_components = 3000)` on train data (`.fit_transform(X_train)`) and test data (`.transform(X_test)`). Then use [GaussianNB](#) to train transformed train data (`X_train_transformed, Y_train`), and make predictions on transformed test data (`.predict(X_test_transformed)`). What is the percentage of instances in test data are correctly classified now?

Provide discussion regarding the performance with trained Naïve Bayes models (including the whole procedure from raw data)